

Diagram 1: CFAR Sliding Window

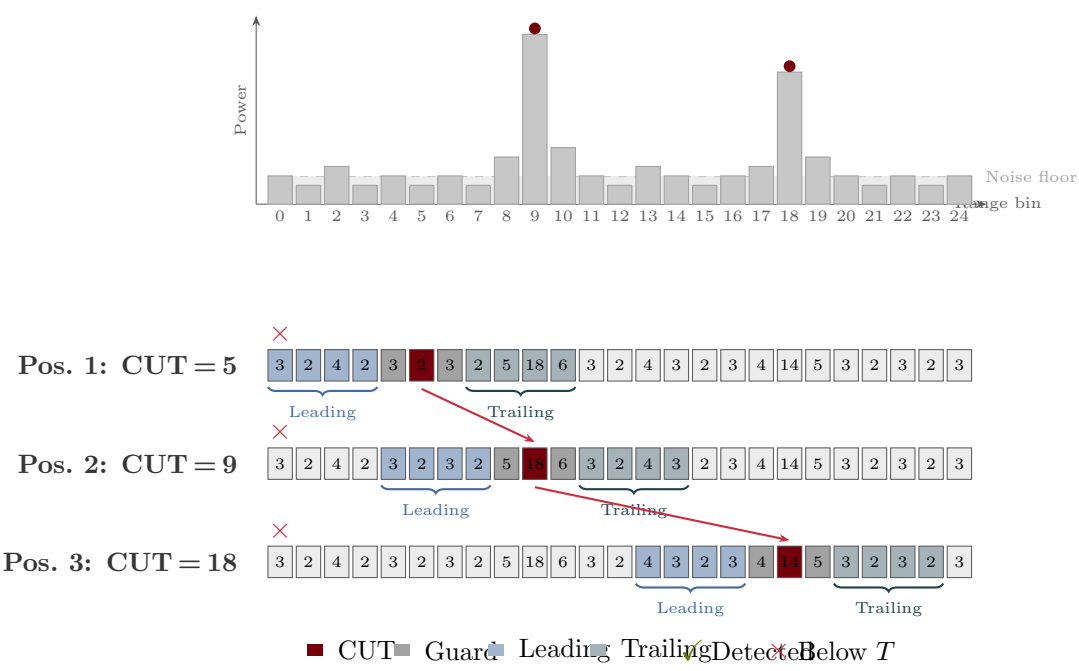


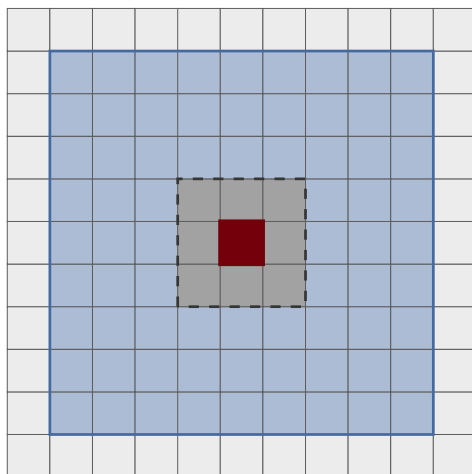
Diagram 1b: CFAR Variant Comparison

Uniform Clutter

2	1	2	1	2	1	2	1	2	1	2
1	2	1	2	2	2	2	2	1	2	1
2	1	2	2	3	3	3	2	2	1	2
1	2	2	3	3	3	3	3	2	2	1
2	2	3	3	3	4	3	3	3	2	2
1	2	3	3	4	15	4	3	3	2	1
2	2	3	3	3	4	3	3	3	2	2
1	2	2	3	3	3	3	3	2	2	1
2	1	2	2	3	3	3	2	2	1	2
1	2	1	2	2	2	2	2	1	2	1
2	1	2	1	2	1	2	1	2	1	2



CA-CFAR



$$Z = \frac{1}{2N} \sum_{i=1}^{2N} x_i$$

■ CUT

■ Guard

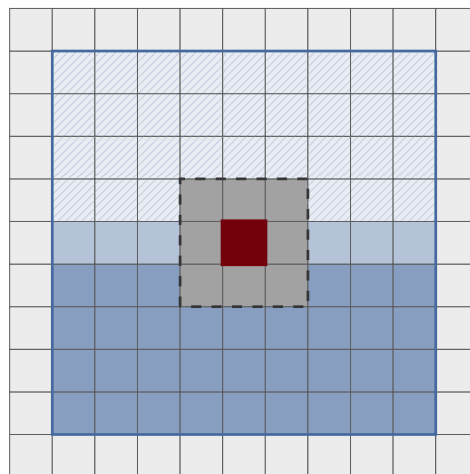
■ Selected Ref.

Strong Leading Edge

1	2	1	1	2	1	2	1	1	2	1
1	1	2	1	1	1	1	1	2	1	1
2	1	1	1	2	2	2	1	1	1	2
1	2	1	2	2	2	2	2	1	2	1
2	1	2	2	3	3	3	2	2	1	2
2	2	2	3	4	15	4	3	2	2	2
3	3	3	3	4	4	4	3	3	3	3
7	8	9	10	8	7	8	10	9	8	7
8	9	12	11	9	8	9	11	12	9	8
9	10	11	10	9	8	9	10	11	10	9
8	9	10	9	8	7	8	9	10	9	8



GO-CFAR



$$Z = \max(\bar{x}_L, \bar{x}_T)$$

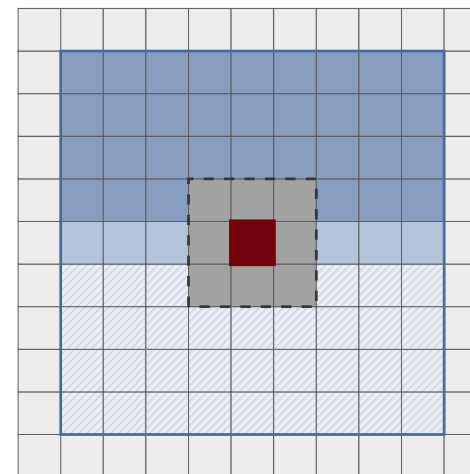
▨ Unused Ref.

Strong Trailing Edge

8	9	10	9	8	7	8	9	10	9	8
9	10	11	10	9	8	9	10	11	10	9
8	9	12	11	9	8	9	11	12	9	8
7	8	9	10	8	7	8	10	9	8	7
3	3	3	3	4	4	4	3	3	3	3
2	2	2	3	4	15	4	3	2	2	2
2	1	2	2	3	3	3	2	2	1	2
1	2	1	2	2	2	2	2	1	2	1
2	1	1	1	2	2	2	1	1	1	2
1	1	2	1	1	1	1	1	2	1	1
1	2	1	1	2	1	2	1	1	2	1



SO-CFAR



$$Z = \min(\bar{x}_L, \bar{x}_T)$$

Diagram 2: Range-Azimuth to PPI Mapping

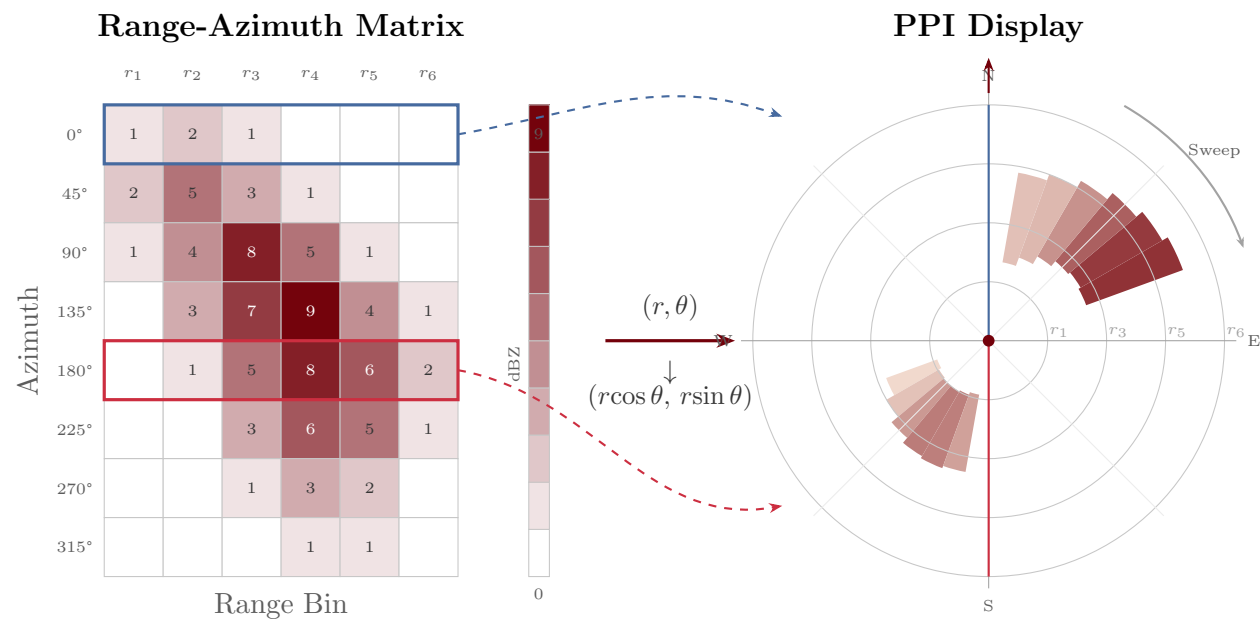


Diagram 3: Coverage Score Illustration

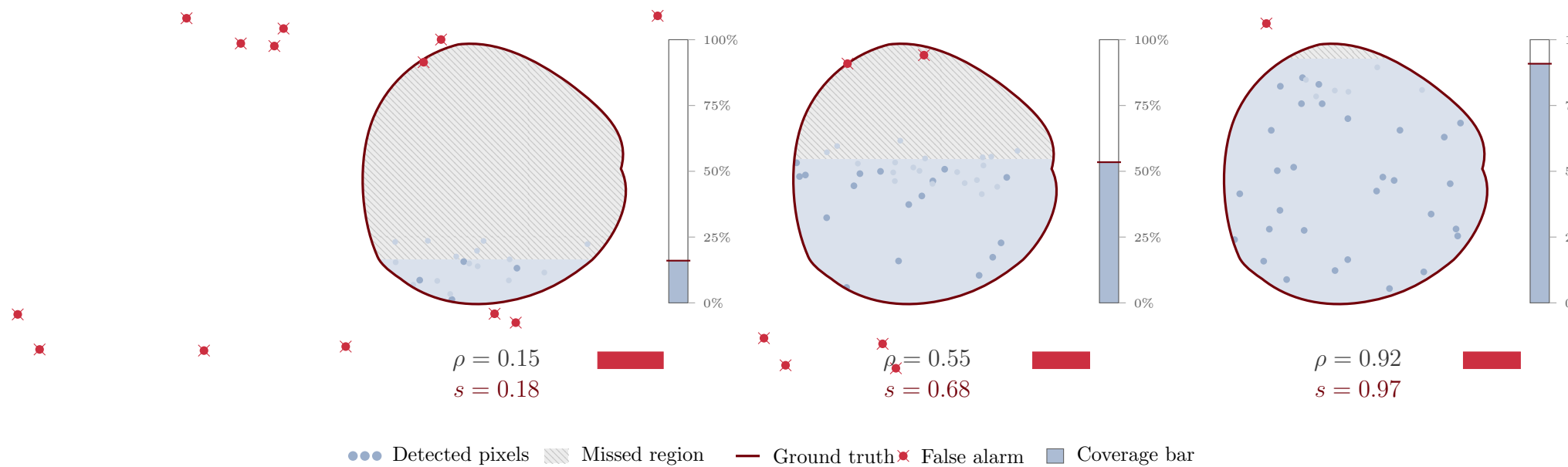


Diagram 4: Guard Cell Size Effect

(a) Guard Too Small

(b) Guard Appropriate

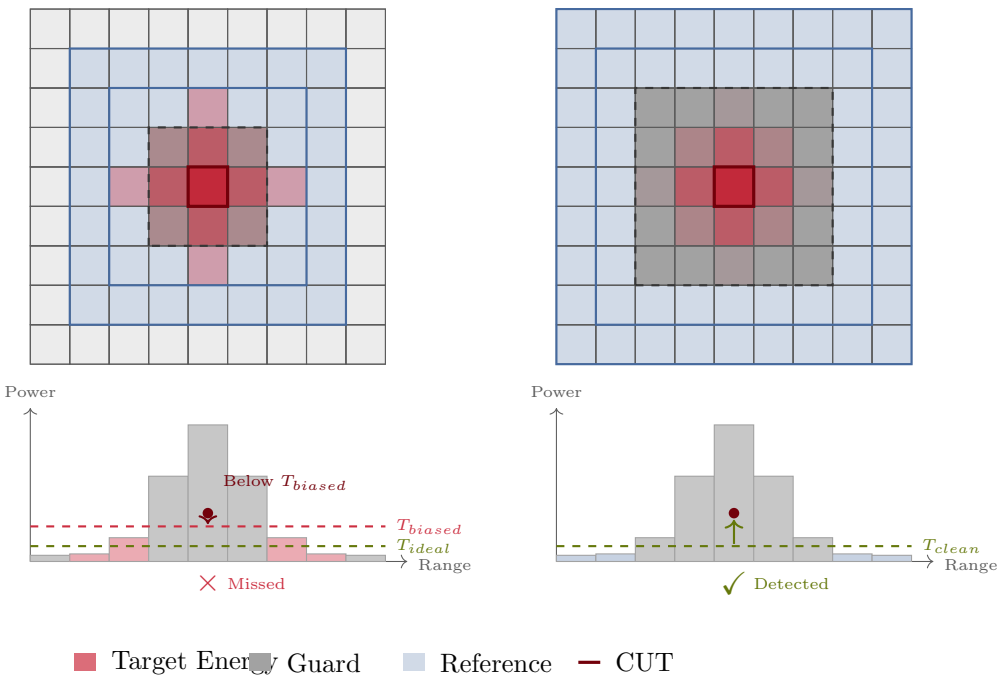


Diagram 5: Integral Image (Summed Area Table)

